

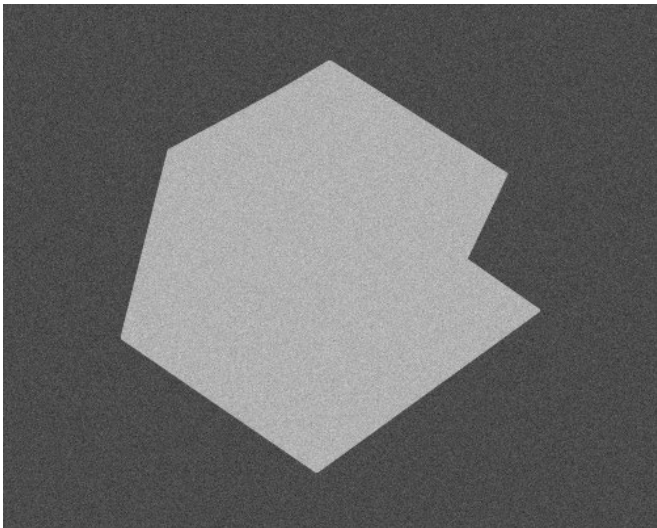
Image Segmentation - 2

Global Thresholding and Statistical Mixture Model

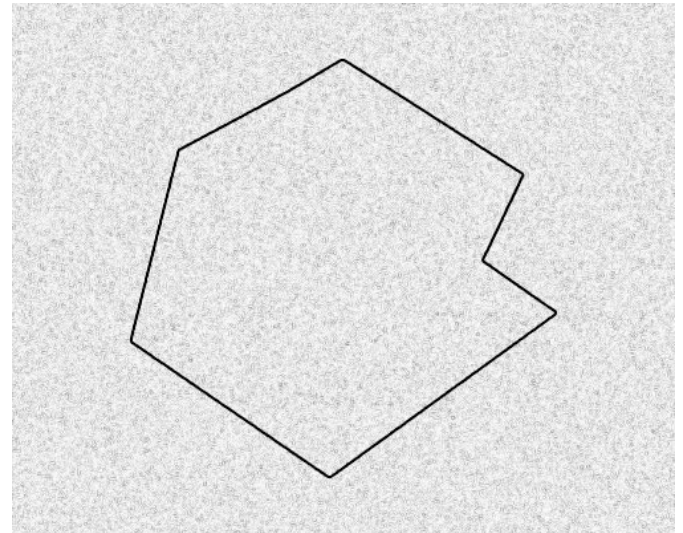
Global intensity-based segmentation (Global thresholding)

Global thresholding

1. Previously, we focused on image segmentation based on the gradient values and boundaries.



Original image

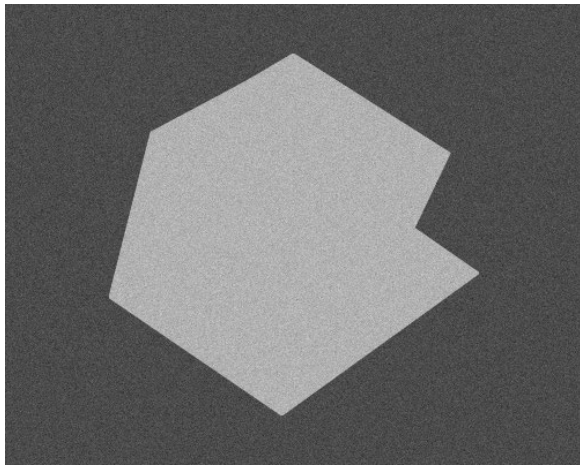


Edge map

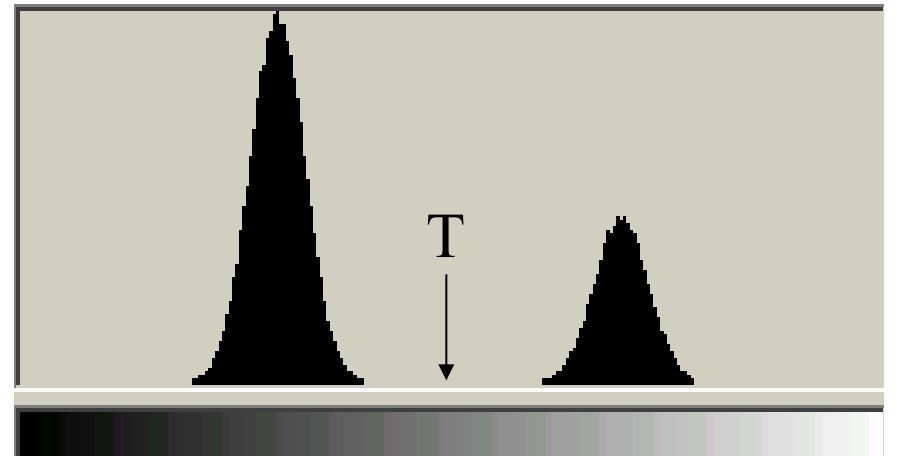
Object can be detected by using edge detection and linking methods

Global thresholding

2. Now, this handout will cover image segmentation by using threshold estimated based on pixel intensity statistics.



Original image

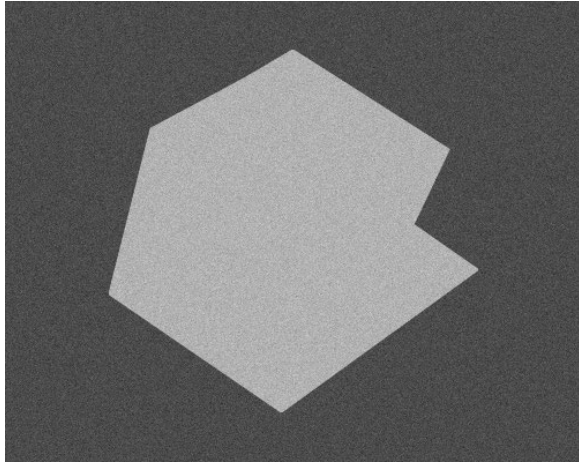


Histogram of the image
(bimodal)

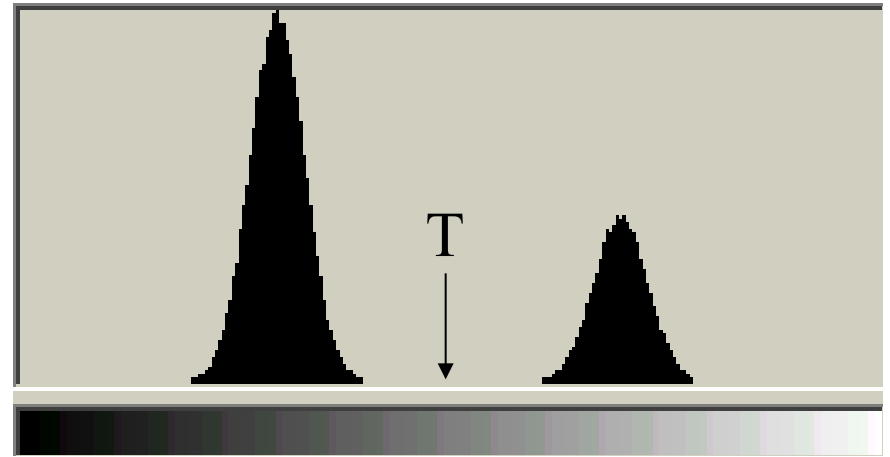
Global thresholding

1. Intensity histogram $h(i)$ = occurrences of pixel intensity value i . For example, $h(i=3) = 100$ means that there are 100 pixels having intensity value $i=3$.
2. For an image $f(x,y)$ having light object and dark background, the histogram consists of two dominant modes (two groups on intensity values).
3. The object is extracted by selecting a threshold T that clearly separates these modes.

Global thresholding



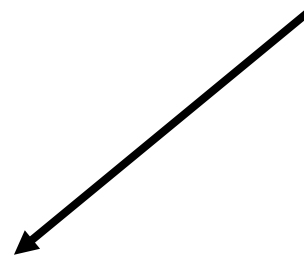
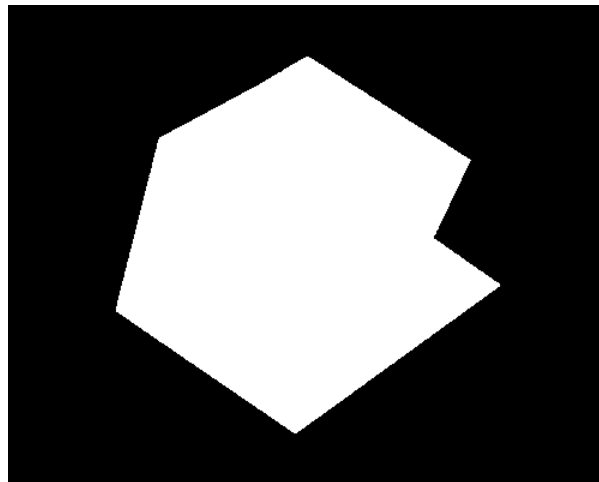
Original image



0

Histogram of the image
(bimodal)

255



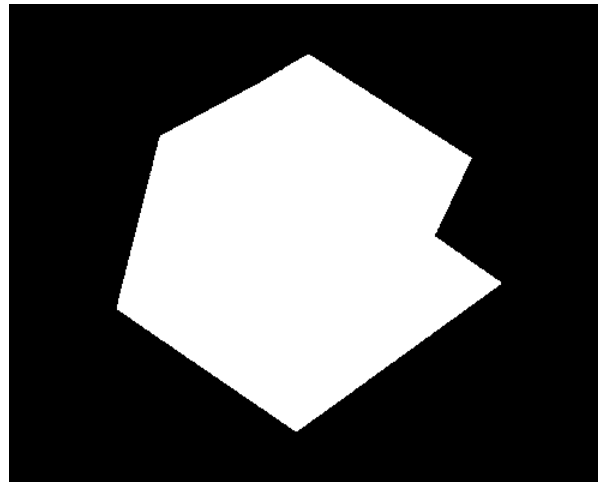
The thresholded (binary) image

Global thresholding

- Each pixel is then labelled according to a pre-defined pixel labelling rule (2-class labelling), which is given by

$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) \geq T \\ 0 & \text{if } f(x, y) < T \end{cases}$$

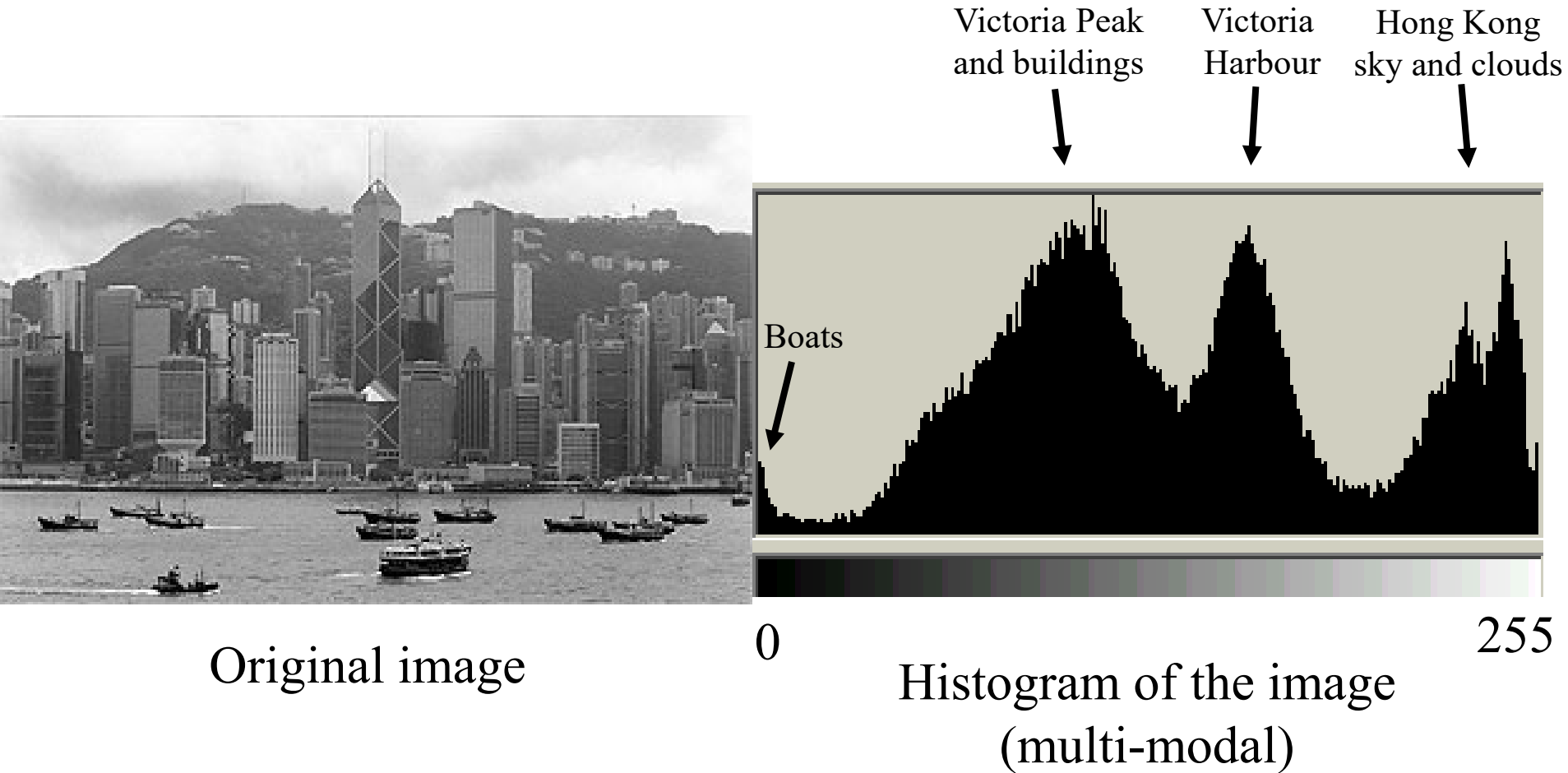
- If $T = 127$,



The thresholded (binary) image

Global thresholding

6. Multi-level thresholding (multi-class labelling)

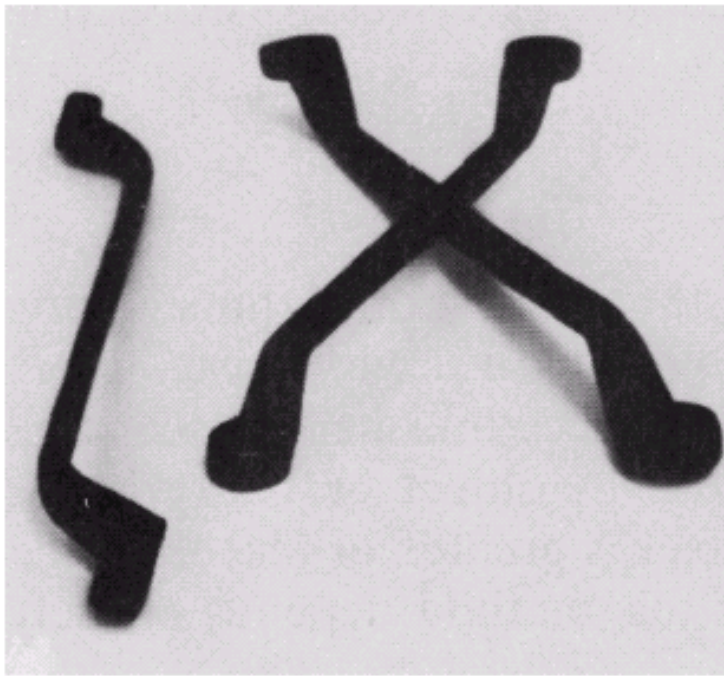


Global thresholding

7. Global vs local/adaptive thresholding
 - a. Global thresholding – threshold does not depend on image locations or local features/neighbouring pixel properties.
 - b. Local thresholding – threshold depends on image locations or local features/neighbouring pixel properties.

Global Thresholding

1. Idea: partitioning the histogram into distinct regions by using a single, global threshold.
2. Segmentation is accomplished by scanning an image pixel by pixel and labeling each pixel as object or background according to a pixel labeling rule.
3. The pixel labeling rule assigns labels to each pixel. The label assignment depends on whether the pixel's grey level is greater or less than the value of T .

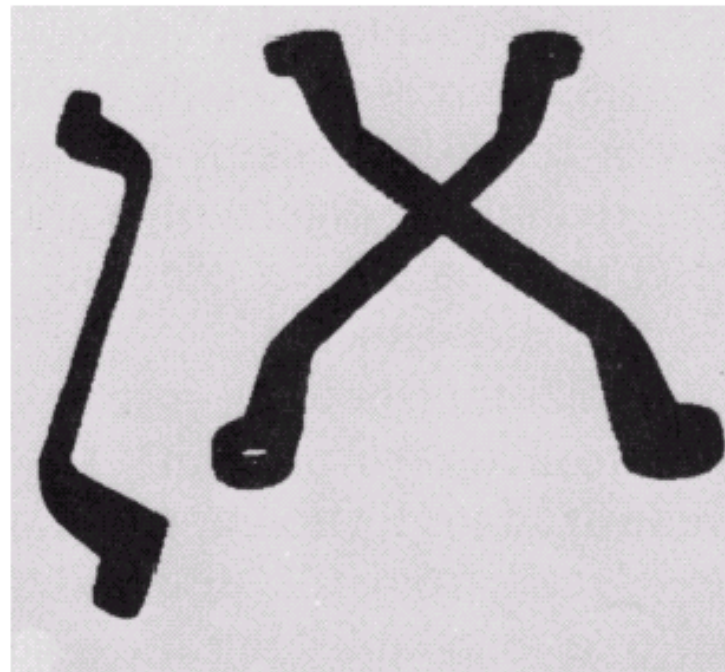


a
b c

FIGURE 10.28
 (a) Original image. (b) Image histogram.
 (c) Result of global thresholding with T midway between the maximum and minimum gray levels.

T = midway between
 max. and min. grey values.

T
 ↓



Global Thresholding

4. How to estimate the global threshold T ?
5. Idea 1: T = midway between max and min grey levels.
6. Idea 2: T = midway between the two peaks of the distributions.
7. Method 1 (automatic, iterative approach with reliance on detecting peaks).
 - a. Select an initial estimate for T .
 - b. Segment the image using T . It creates two regions G_1 (all pixel intensity values $> T$) and G_2 (otherwise).
 - c. Compute the average gray level values (means) μ_1 and μ_2 for the pixels in regions G_1 and G_2 , respectively.
 - d. Compute a new threshold value

$$T = \frac{\mu_1 + \mu_2}{2}$$

- e. Repeat steps b through d until the change in T is small.

Global Thresholding

8. This iteration can converge to the threshold.
9. This final threshold minimizes a square-error value, which is the total squared intensity differences between the gray level values and the mean of the corresponding region.
10. Therefore, it has the smallest total intensity variation (total squared intensity differences) of all regions.

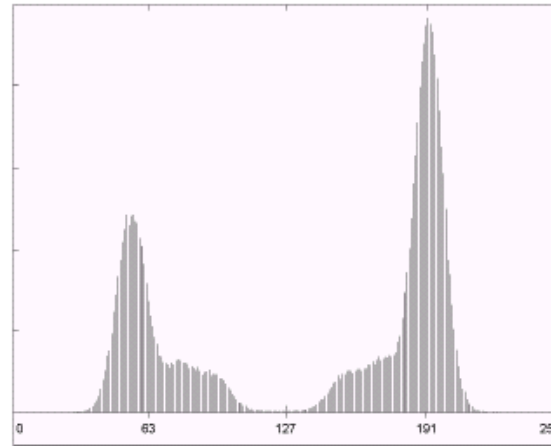
$$e^2 = \int_0^T (i - A)^2 h(i) di + \int_T^N (B - i)^2 h(i) di,$$

where

T = Threshold

A and B are the means of the regions separated by T

$h(i)$ = intensity histogram



a b
c

FIGURE 10.29

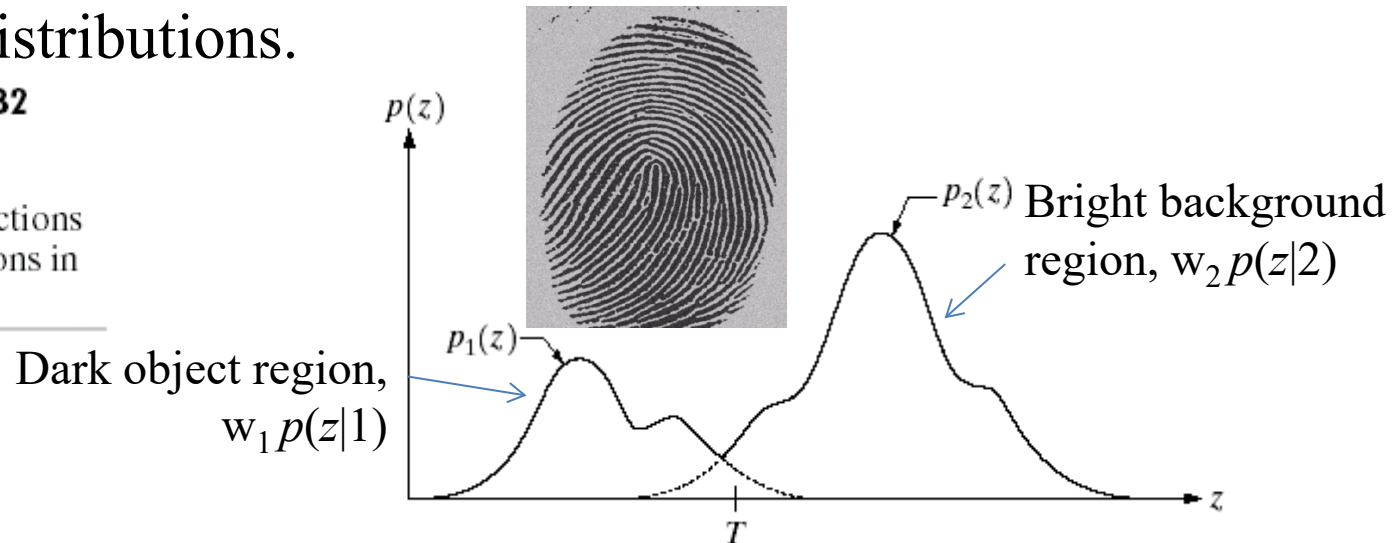
(a) Original image. (b) Image histogram. (c) Result of segmentation with the threshold estimated by iteration. (Original courtesy of the National Institute of Standards and Technology.)



Global Thresholding

8. Method 2 (automatic, statistical, minimum error approach).
- a. We assume that the image histogram is a mixture of two distributions.

FIGURE 10.32
Gray-level probability density functions of two regions in an image.



- b. The probabilities $p(z|1)$ and $p(z|2)$ can be the Gaussian distributions, Rayleigh or others.
- c. The mixture model is defined as

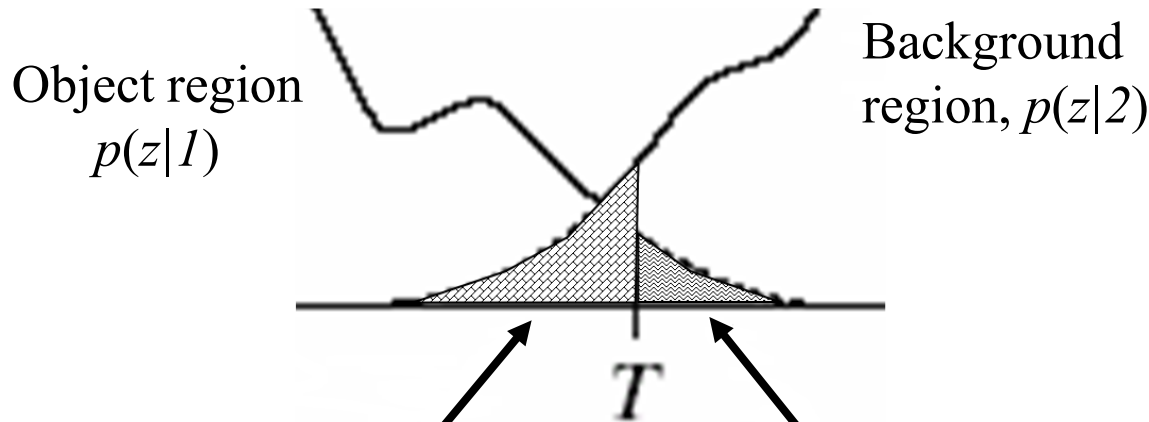
$$p(z) = w_1p(z|1) + w_2p(z|2)$$

where $w_1 + w_2 = 1$, and $0 \leq w_1, w_2 \leq 1$

Global Thresholding

8. Method 2 (automatic, statistical, minimum error approach).

d. Error is defined as



$$E(T) = \int_{-\infty}^T w_2 p(z|2) dz + \int_T^{\infty} w_1 p(z|1) dz$$

Misclassifying background pixels into object pixels. Misclassifying object pixels into background pixels.

e. By minimising $E(T)$ with respect to T , we get

$$\text{when } w_1 p(T|1) = w_2 p(T|2)$$

$E(T)$ is minimum.

Basic Global Thresholding

8. Method 2 (automatic, statistical, minimum error approach).

f. If $p(z|1)$ and $p(z|2)$ are Gaussian distributed,

$$p(z) = \frac{w_1}{\sqrt{2\pi}\sigma} e^{\frac{-(z-\mu_1)^2}{2\sigma^2}} + \frac{w_2}{\sqrt{2\pi}\sigma} e^{\frac{-(z-\mu_2)^2}{2\sigma^2}}$$

then the minimum error threshold is given by

$$T = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma^2}{\mu_1 - \mu_2} \ln \left(\frac{w_2}{w_1} \right)$$

Given the observed histogram and statistical mixture model,

how to estimate the parameters (w_1 , w_2 , means and SDs)?

Answer: Expectation-Maximization (EM) method can be used.

https://en.wikipedia.org/wiki/Expectation%E2%80%93maximization_algorithm

Statistical mixture models

1. The probability density function (PDF) is formed from a linear combination of M basis functions (e.g., Gaussian distribution).
2. Mixture distribution (or PDF) is given by

$$p(z) = \sum_{j=1}^M p(z | j) p(j)$$

where M represents the number of basis functions (classes)

$p(z | j)$ represents basis function

$p(j) = w_j$ represents mixing parameter

Related reference:

<http://www.stat.cmu.edu/~cshalizi/uADA/12/lectures/ch20.pdf>

Or, search “Gaussian Mixture Model” on the Internet for more information.

Statistical mixture models

3. Constraints

$$(1) \quad \sum_{j=1}^M p(j) = 1$$

$$(2) \quad 0 \leq p(j) \leq 1$$

$$(3) \quad \int p(z | j) dx = 1$$

$$\text{e.g., } p(z | j) = \frac{1}{\sqrt{2\pi\sigma_j^2}} \exp\left[\frac{-(z - \mu_j)^2}{2\sigma_j^2}\right]$$

Gaussian distribution

Statistical mixture models

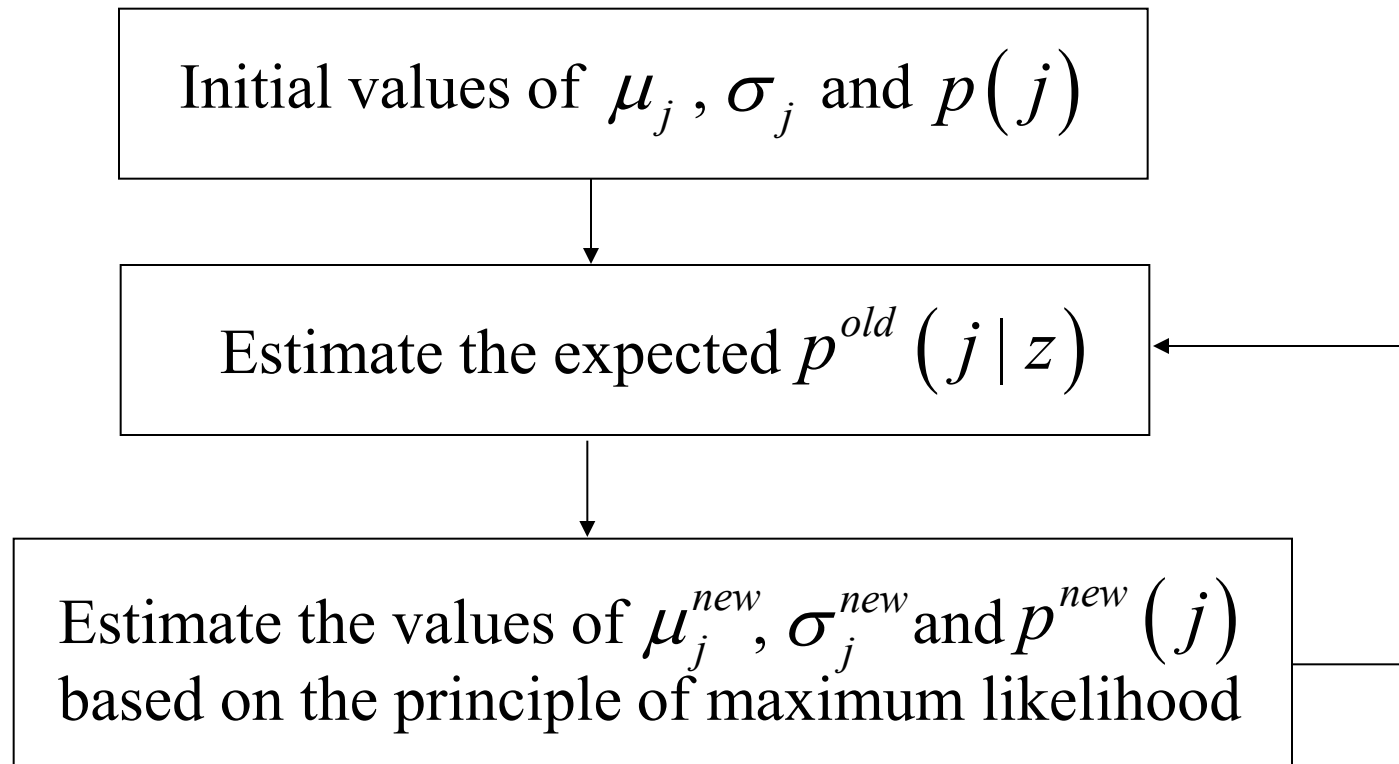
4. Posterior probability

$$p(j | z) = \frac{p(z | j) p(j)}{p(z)}$$

$$\sum_{j=1}^M p(j | z) = 1$$

Expectation-Maximization (EM) method

1. The EM method is an iterative scheme for finding the values of the parameters in a mixture model. The concept is to maximize the expected negative log-likelihood function $E = -\ln L$ of an observed image.



Expectation-Maximization (EM) method

2. Update equations are shown below. Let N be the number of pixels in an image. With pixel index $n = 1, \dots, N$, z^n represents pixel intensity at the n^{th} pixel.
3. Iteration continues until convergence is reached.

$$\mu_j^{new} = \frac{\sum_n p^{old} (j | z^n) z^n}{\sum_n p^{old} (j | z^n)}$$

$$(\sigma_j^{new})^2 = \frac{\sum_n p^{old} (j | z^n) (z^n - \mu_j^{new})^2}{\sum_n p^{old} (j | z^n)}$$

$$p(j)^{new} = \frac{1}{N} \sum_n p^{old} (j | z^n)$$

Otsu's method

- The method is designed for finding the optimum global threshold for image binarization.
- The method is optimum in the sense that it maximizes the between-class variance.
- There are six steps.
- Step 1: Compute the normalized histogram of the input image. Denote the components of the histogram by

$$p_i = \frac{n_i}{MN}, \quad i = 0, 1, 2, \dots, L-1$$

where

L = number of gray levels

n_i = number of pixels with intensity i

M = number of rows

N = number of columns

Otsu's method

- Step 2: Compute the cumulative sums

$P_1(k)$, $k = 0, 1, 2, \dots, L-1$ where L =number of gray levels
 k = current threshold for
thresholding the input image into
two classes C_1 and C_2 .

The probability that a pixel is assigned to class C_1 = $P_1(k) = \sum_{i=0}^k p_i$

- Step 3: Compute the cumulative means

$$m(k) = \sum_{i=0}^k ip_i, \quad k = 0, 1, 2, \dots, L-1$$

Otsu's method

- Step 4: Compute the global intensity mean

$$m_G = \sum_{i=0}^{L-1} ip_i \quad \text{where } L = \text{number of gray levels}$$

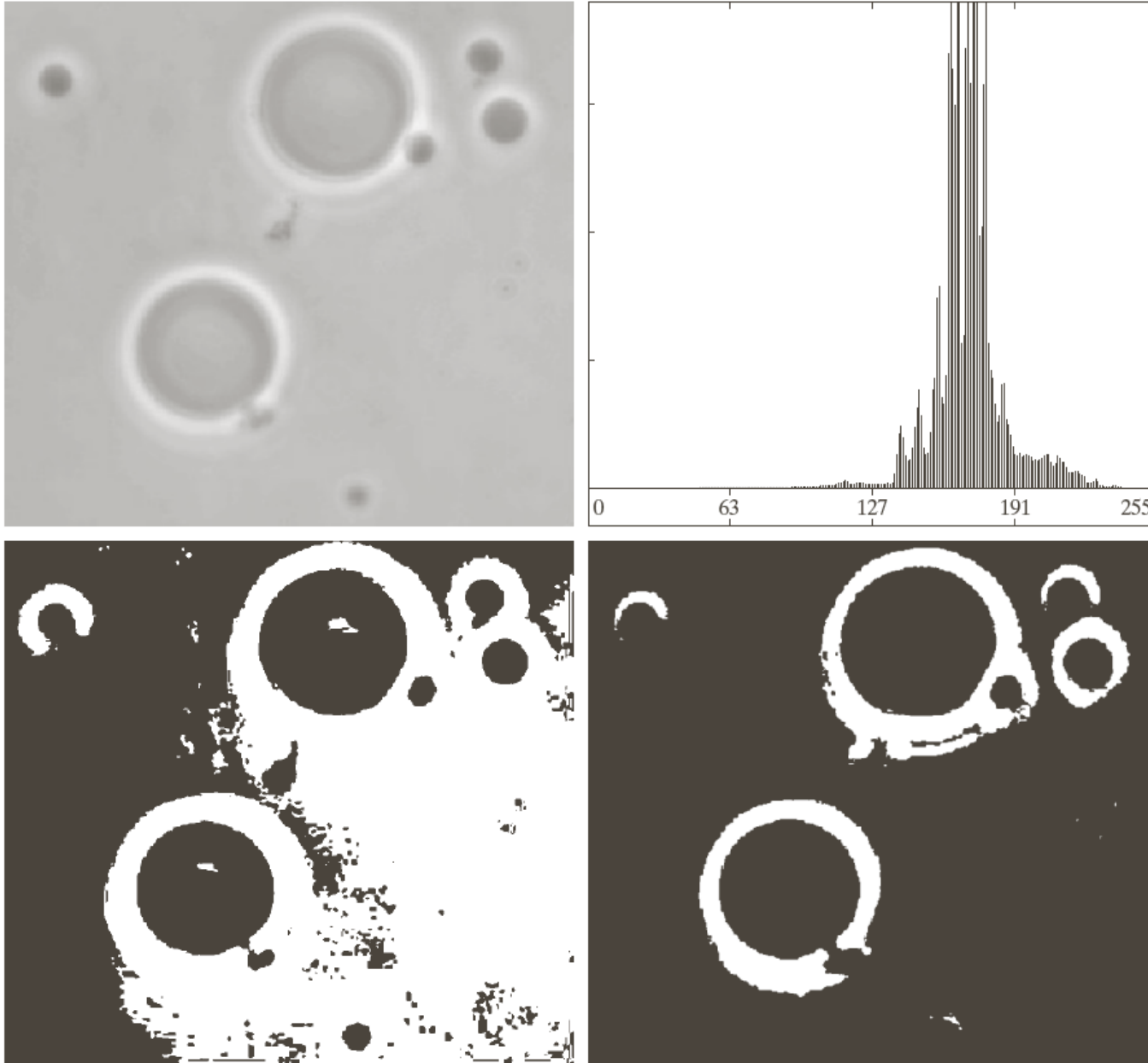
- Step 5: Compute the between-class variance

$$\sigma_B^2(k) = \frac{[m_G P_1(k) - m(k)]^2}{P_1(k)[1 - P_1(k)]}, \quad k = 0, 1, 2, \dots, L-1$$

- Step 6: Obtain the Otsu threshold, k^* , as the value for k for which the value of between-class variance is maximum. If the maximum is not unique, obtain k^* by averaging the values of k corresponding to the various maxima detected.

$$\sigma_B^2(k^*) = \max_{0 \leq k \leq L-2} \sigma_B^2(k)$$

Otsu's method

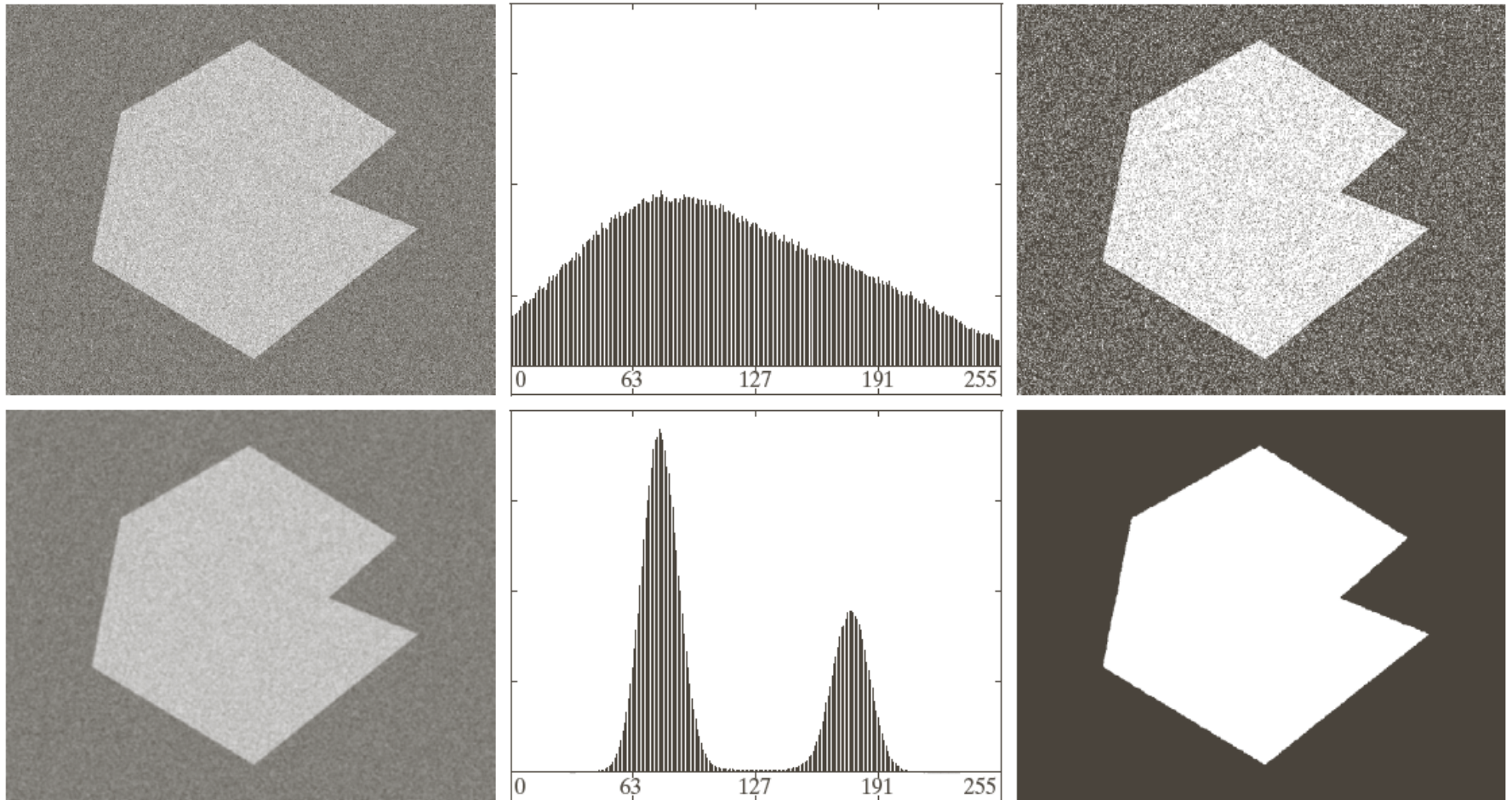


a	b
c	d

FIGURE 10.39

(a) Original image.
(b) Histogram (high peaks were clipped to highlight details in the lower values).
(c) Segmentation result using the basic global algorithm from Section 10.3.2.
(d) Result obtained using Otsu's method. (Original image courtesy of Professor Daniel A. Hammer, the University of Pennsylvania.)

Otsu's method



a	b	c
d	e	f

FIGURE 10.40 (a) Noisy image from Fig. 10.36 and (b) its histogram. (c) Result obtained using Otsu's method. (d) Noisy image smoothed using a 5×5 averaging mask and (e) its histogram. (f) Result of thresholding using Otsu's method.

The figure is provided by Pearson Education, Digital Image Processing, Gonzalez & Woods, www.ImageProcessingPlace.com